

SUNCLAVE · DESIGN DECISION RECORD

SunClave recommendations accepted

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DOCUMENT CONTROL

Document ID	SCL-DDR-002	Version	0.2
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/sunclave		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	Items 5 and 12 decided by Amish on 2026-10-02 (recommendations approved, SCL-DEC-001)
0.1	2026-09-25	Amish Chadha	Record Amish's acceptance of all open recommendations (items 13 to 18) and the items that remain open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted (items 13 to 18); items 5 and 12 were decided on 2026-10-02 (SCL-DEC-001)

Safety: SunClave is a research and educational prototype, not a medical device. It concentrates sunlight to a burning intensity and holds steam at about 121 °C (250 °F) in a pressure vessel. Item 5 (mounting the lid fittings) is a pressure-safety trade-off with no recommendation and stays open.

■ Context

After the TRL 3 session, SCL-DDR-001 and `docs/REVIEW.md` listed items 5 and 12 to 18 as "Proposed, awaiting Amish". Items 13 to 18 each carried a recommendation; items 5 and 12 did not. On 2026-09-25 Amish wrote, in chat: "i accept all your recommendations, go with them across all repos."

This record lists what that instruction decides, what changed in the repo because of it, and what stays open. TRL 4 remains on hold by Amish's instruction.

■ Options considered

The options for each item are in SCL-DDR-001 Table 2 and `docs/REVIEW.md` (TRL 3 section). Where an item offered several options, the recommended option is the decision.

■ Decision

Table 1. Newly decided items. Each is "Decided by Amish, 2026-09-25: go with recommendation".

#	Item	Decision	What changed in the repo
13	R14 tracking	Option (a): relax R14 to retargeting no more often than every 12 min, with a logger reminder at each interval	SCL-REQ-001 v0.4 R14 relaxed (15 min to 12 min); SCL-CAL-001 v0.2 retargets every 12 min, so the mean absorbed power rises from 630 W to 638 W and the central cycle falls from 79.9 min to 79.0 min; R14 moves from not met to met (about 13 min held against 12 min). The reminder is a firmware rule for the existing buzzer (BOM item 14); no firmware is written
14	R16 mass	Option (a): relax the total to 45 kg and keep the 20 kg piece limit	SCL-REQ-001 v0.4 R16 relaxed (40 kg to 45 kg); R16 moves from not met to met (41.0 kg; largest piece 19.4 kg); drawing SCL-DWG-001 note updated
15	R11 guard and hot surfaces	Option (a): restate R11 so the whole pressure vessel and its fittings are a hot zone, reached only with the dish turned off the sun, relying on the parking cover, goggles and a marked keep-out on the ground; no physical guard	SCL-REQ-001 v0.4 R11 restated; new BOM item 21, keep-out marking (\$6); SCL-PRC-001 v0.4 safety section; drawing note; R11 moves from not met to met by design review
16	R15 stability	Option (a): four locking castors instead of two, plus a rule to park the dish face-up in high wind	BOM item 4 (\$40 to \$44); <code>cad/src/model.py</code> adds a brake pedal on each castor (<code>CASTOR_LOCKS = 4</code>), STEP and STL re-exported; drawing SCL-DWG-001 Rev P1 to P2; grip rises from 109 N to 218 N against 153 N of wind. The tipping factor stays 1.30, so R15 stays at risk
17	R10 water at altitude	Option (a): an operating rule to trim the dish off the sun during holds so only a little steam vents, with a water-budget (trim) reminder from the logger	SCL-REQ-001 v0.4 R10 restated to include the rule; the logger shows a trim reminder whenever the computed hold exceeds 40 min; SCL-CAL-001 v0.2 shows 1.31 kg left at 1,800 m and 1.28 kg at 2,400 m in 1,000 W/m ² sun with trimming (0.14 kg at 1,800 m untrimmed); R10 moves from at risk to met
18	R12 transducer span	Adopt a 0 to 300 kPa absolute transducer instead of 0 to 500 kPa, with an overpressure rating above the relief set point	BOM item 14 spec (overpressure 600 kPa or more, same \$58); SCL-CAL-001 v0.2 check uncertainty 0.89 K to 0.64 K and pressure accuracy 5 kPa to 3 kPa; R12 moves from at risk to met, narrowly (32 % of the 2 K threshold)

The budget is unchanged at \$450 (`project.yaml`). Parts rise from \$430 to \$440 (castors \$4, keep-out marking \$6), a margin of \$10.

Items that remain open

Table 2. Items with no recommendation here, decided on 2026-10-02.

#	Item	Status
5	Mounting the gauge, relief valve and probe gland: drill the maker's lid, or a cooker with factory ports or an adapter plate on the regulator stem	Decided by Amish, 2026-10-02 (recommendation approved, SCL-DEC-001): the maker's lid is not drilled; a pressure canner sold with a factory gauge and relief valve is used, and the probe gland goes on an adapter plate on the vent stem that keeps a bore of 3 mm or more and leaves the overpressure plug untouched; a lid is drilled only with the maker's written approval.
12	First partner and site for co-design	Decided by Amish, 2026-10-02 (recommendation approved, SCL-DEC-001): first partner to approach is a university biomedical or global health engineering group with an established rural clinic partner in a sunny region below 2,400 m, for example through an Engineering World Health university chapter.

Cross-repo actions

None. No decision in this record needs another repo to change.

■ Consequences

- SCL-PRC-001 and SCL-REQ-001 move to v0.4, SCL-CAL-001 to v0.2, SCL-DDR-001 to v0.2 and drawing SCL-DWG-001 to Rev P2.
- Requirement status (SCL-CAL-001 v0.2): 14 met, 2 at risk (R4 cycle time, R15 stability), 1 not verifiable at TRL 3 (R1), 0 not met. Before this record: 9 met, 4 at risk, 1 not verifiable, 3 not met. R11, R14 and R16 are met because their targets were restated or relaxed, not because the design improved.
- The retarget and trim reminders are recorded as firmware rules only. Writing and testing firmware, buying parts and building anything are TRL 4 work, which is on hold by Amish's instruction.
- `trl` and `trl_target` stay at 3.